Article

Full title of the paper (Capitalized)

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Simple Summary: A Simple summary goes here.

Abstract: A single paragraph of about 200 words maximum. For research articles, abstracts should give a pertinent overview of the work. We strongly encourage authors to use the following style of structured abstracts, but without The abstract should be an objective representation of the article, it must not contain results which are not presented and substantiated in the main text and should not exaggerate the main conclusions. headings: 1) Background: Place the question addressed in a broad context and highlight the purpose of the study; 2) Methods: Describe briefly the main methods or treatments applied; 3) Results: Summarize the article's main findings; and 4) Conclusion: Indicate the main conclusions or interpretations.

Keywords: keyword 1; keyword 2; keyword 3 (list three to ten pertinent keywords specific to the article, yet reasonably common within the subject discipline.).

1. Version

This Rmd-skeleton uses the mdpi Latex template published 2023-03-25. However, the official template gets more frequently updated than the **rticles** package. Therefore, please make sure prior to paper submission, that you're using the most recent .cls, .tex and .bst files (available here).

2. Article Header Information

The YAML header includes information needed mainly for formatting the front and back matter of the article. Required elements include:

```
title: Title of the paper
author:
  - name: first and last name
    affil: |
      One or more comma seperated numbers corresponding to affilitation
      and one or more  comma seperated symbols corresponding
      optional notes.
    orcid: optional orcid number
affiliation:
  - num: 1,..., n for each affiliation
    address: required
    email: required
authorcitation: |
  Lastname, F.
correspondence: |
  email@email.com; Tel.: +XX-000-00-0000.
journal: notspecified
type: article
status: submit
```

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Table 1. MDPI article types.

abstract	entry	retraction
addendum	expressionofconcern	review
article	extendedabstract	perspective
book	datadescriptor	protocol
bookreview	editorial	shortnote
briefreport	essay	studyprotocol
casereport	erratum	systematicreview
comment	hypothesis	supfile
commentary	interestingimage	technicalnote
communication	obituary	viewpoint
conferenceproceedings	opinion	guidelines
correction	projectreport	registeredreport
conferencereport	reply	tutorial

Journal options are in Table 2. The `status` variable should generally not be changed by authors. The `type` variable describes the type of of submission and defaults to `article` but can be replaced with any of the ones in Table 1

2.1. Journal Specific YAML variables

```
# for journal Diversity,
# add the Life Science Identifier using:
lsid: http://zoobank.org/urn:lsid:zoobank.org:act:nnnn


# for journal Applied Sciences
# add featured application
featuredapplication: |
  Authors are encouraged to provide a concise
  description of the specific application or
  a potential application of the work. This
  section is not mandatory.

# for the journal Data
# add dataset doi and license
dataset: https://doi.org/10.1000/182
datasetlicense: CC-BY-4.0

# for the journal Toxins
# add key contributions
keycontributions: |
  The breakthroughs or highlights of the manuscript.
  Authors can write one or two sentences to describe
  the most important part of the paper.

# for the journal Encyclopedia
encyclopediadef: |
  For entry manuscripts only: please provide a brief overview
  of the entry title instead of an abstract.
entrylink: The Link to this entry published on the encyclopedia platform.

# for the journal Advances in Respiratory Medicine
```

```
# add highlights
addhighlights: |
  This is an obligatory section in ‘‘Advances in Respiratory Medicine’’,
  whose goal is to increase the discoverability and readability of the
  article via search engines and other scholars. Highlights should not
  be a copy of the abstract, but a simple text allowing the reader to
  quickly and simplified find out what the article is about and what can
  be cited from it. Each of these parts should be devoted up to 2~bullet
  points.
```

Table 2. MDPI journal names.

acoustics	biomedinformatics	dentistry	galaxies	jcp	metabolites	philosophies	socsci
actuators	biomimetics	dermato	games	jcs	metals	photochem	software
addictions	biomolecules	dermatopathology	gases	jcto	meteorology	photonics	soilsystems
admsci	biophysica	designs	gastroent	jdb	methane	phycology	solar
adolescents	biosensors	devices	gastrointestdisord	jeta	metrology	physchem	solids
aerobiology	biotech	diabetology	gels	jfb	micro	physics	spectroscj
aerospace	birds	diagnostics	genealogy	jfmk	microarrays	physiologia	sports
agriculture	bloods	dietetics	genes	jimaging	microbiolres	plants	standards
agriengineering	blsf	digital	geographies	jintelligence	micromachines	plasma	stats
agrochemicals	brainsci	disabilities	geohazards	jlpea	microorganisms	platforms	std
agronomy	breath	diseases	geomatics	jmmp	microplastics	pollutants	stresses
ai	buildings	diversity	geosciences	jmp	minerals	polymers	surfaces
air	businesses	dna	geotechnics	jmse	mining	polysaccharides	surgeries
algorithms	cancers	drones	geriatrics	jne	modelling	poultry	suschem
allergies	carbon	dynamics	grasses	jnt	molbank	powders	sustainability
alloys	cardiogenetics	earth	gucdd	jof	molecules	preprints	symmetry
analytica	catalysts	ebj	hazardousmatters	joitmc	mps	proceedings	synbio
analytics	cells	ecologies	healthcare	jor	msf	processes	systems
anatomia	ceramics	econometrics	hearts	journalmedia	mti	prosthesis	targets
animals	challenges	economies	hemato	jox	muscles	proteomes	taxonomy
antibiotics	chemengineering	education	hematolrep	jpm	nanoenergyadv	psf	technologies
antibodies	chemistry	ejihpe	heritage	jrfm	nanomanufacturing	psych	telecom
antioxidants	chemosensors	electricity	higheredu	jsan	nanomaterials	psychiatryint	test
applbiosci	chemproc	electrochem	highthroughput	jtaer	ncrna	psychoactives	textiles
appliedchem	children	electronicmat	histories	jvd	ndt	publications	thalassrep
appliedmath	chips	electronics	horticulturae	jzbg	network	quantumrep	thermo
applmech	cimb	encyclopedia	hospitals	kidneydial	neuroglia	quaternary	tomography
applmicrobiol	civileng	endocrines	humanities	kinasesphosphatases	neurolint	qubs	tourismhosp
applnano	cleantechnol	energies	humans	knowledge	neurosci	radiation	toxics
applsci	climate	eng	hydrobiology	land	nitrogen	reactions	toxins
aquacj	clinpract	engproc	hydrogen	languages	notspecified	receptors	transplantology

architecture
arm
arthropoda
arts

asc
asi
astronomy
atmosphere
atoms

audiolres
automation
axioms
bacteria
batteries

bdcc
behavsci
beverages
biochem
bioengineering

biologics
biology
biomass
biomechanics
biomed

biomedicines

clockssleep
cmd
coasts
coatings

colloids
colorants
commodities
compounds
computation

computers
condensedmatter
conservation
constrmater
cosmetics

covid
crops
cryptography
crystals
csmf

ctn
curroncol
cyber
dairy
data

ddc

entomology
entropy
environments
environsciproc

epidemiologia
epigenomes
est
fermentation
fibers

fintech
fire
fishes
fluids
foods

forecasting
forensicsci
forests
foundations
fractalfract

fuels
future
futureinternet
futurepharmacol
futurephys

futuretransp

hydrology
hygiene
idr
ijerph

ijfs
ijgi
ijms
ijns
ijpb

ijtm
ijtpp
ime
immuno
informatics

information
infrastructures
inorganics
insects
instruments

inventions
iot
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jcdd

jcm

laws
life
liquids
literature

livers
logics
logistics
lubricants
lymphatics

machines
macromol
magnetism
magnetochemistry
make

marinedrugs
materials
materproc
mathematics
mca

measurements
medicina
medicines
medsci
membranes

merits

nri
nursrep
nutraceuticals
nutrients

obesities
oceans
ohbm
onco
oncopathology

optics
oral
organics
organoids
osteology

oxygen
parasites
parasitologia
particles
pathogens

pathophysiology
pediatrrep
pharmaceuticals
pharmaceutics
pharmacoepidemiology

pharmacy

recycling
regeneration
religions
remotesensing

reports
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resources
rheumato
risks

robotics
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safety
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scipharm

sclerosis
seeds
sensors
separations
sexes

signals
sinusitis
skins
smartcities
sna

societies

transportation
traumacare
traumas
tropicalmed

universe
urbansci
uro
vaccines
vehicles

venereology
vetsci
vibration
virtualworlds
viruses

vision
waste
water
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wevj

wind
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3. Introduction

The introduction should briefly place the study in a broad context and highlight why it is important. It should define the purpose of the work and its significance. The current state of the research field should be reviewed carefully and key publications cited. Please highlight controversial and diverging hypotheses when necessary. Finally, briefly mention the main aim of the work and highlight the principal conclusions. As far as possible, please keep the introduction comprehensible to scientists outside your particular field of research. Citing a journal paper [1,2]. And now citing a book reference Gujer [3]. Some MDPI journals use Chicago and others use APA, this template should choose the correct citation format for you once you specify the journal in the YAML header.

To use endnotes, change `endnotes: true` in the YAML header, then use `\endnote{This is an endnote.}`.

###Preguntas a responder que he pensado (Javi):

1. Oferta, Demanda y Capacidad Ciclista • ¿Existe correspondencia entre la tipología de aparcamiento y el número de plazas? Analizar la distribución del número de plazas por punto de estacionamiento (plazas), identificando zonas con alta atomización (muchos puntos de pocas plazas) frente a zonas con macroaparcamientos. • ¿Dónde se concentran los "desiertos" de estacionamiento ciclista? Calcular la distancia euclídea o de red de cada barrio/sección censal al aparcamiento de bicicletas más cercano para detectar bolsas urbanas desatendidas. • ¿Cómo interactúan los aparcamientos públicos en horquilla/U invertida con las estaciones de Valenbisi? Evaluar si la infraestructura para bicicleta privada complementa a las estaciones de bicicleta compartida o si compiten en las mismas localizaciones centrales dejando la periferia desabastecida.

2. Intermodalidad con el Transporte Público (EMT y Metro) • ¿Qué paradas de autobús (EMT) o estaciones de metro actúan como nodos intermodales reales? Calcular mediante buffers de proximidad (por ejemplo, radios caminables de 50 a 100 metros) qué porcentaje de paradas de autobús disponen de aparcamiento de bicis en su entorno inmediato.

   • ¿Las líneas de EMT con mayor demanda cuentan con suficiente infraestructura ciclista en sus cabeceras y nodos clave? Identificar puntos críticos de conexión periférica donde los usuarios podrían cambiar de la bicicleta al autobús urbano.

3. Conexión con la Infraestructura Ciclista (Carril Bici) • ¿Están los aparcamientos ubicados a lo largo de la red de itinerarios ciclistas o exigen circular por calzada/acera?

   Calcular la distancia de cada aparcamiento al eje de carril bici más próximo para detectar paradas aisladas que carecen de acceso seguro. • ¿Cuál es la densidad de aparcamientos por kilómetro de carril bici en los distintos distritos?

4. Relación Sociodemográfica y Territorial (Barrios y Población) • ¿Existe equidad espacial en el ratio de plazas de bici por habitante? Cruzar los límites de distritos/barrios y el padrón de habitantes para calcular:

$$\text{Ratio} = \frac{\text{Plazas de aparcamiento de bicis}}{\text{Población total o población joven (16-35 años)}}$$

   • ¿Se correlaciona la densidad de aparcamientos con la renta media o la densidad poblacional? Verificar si los barrios con mayor densidad residencial o menor renta disponen de una proporción equivalente de infraestructura ciclista frente al centro histórico y zonas turísticas/comerciales.

5. Puntos de Interés (Equipamientos Públicos) • ¿Tienen los centros educativos (universidades, institutos) y centros sanitarios suficiente cobertura de plazas? Medir la concentración de plazas disponibles en un radio de 200 m de campus universitarios

(Tarongers, Blasco Ibáñez, Burjassot/UPV) y hospitales frente a zonas puramente residenciales.

###Javi:
Cargamos las bibliotecas que vamos viendo que hagan falta

```
library(sf)
```

4. Materials and Methods

Materials and Methods should be described with sufficient details to allow others to replicate and build on published results. Please note that publication of your manuscript implicates that you must make all materials, data, computer code, and protocols associated with the publication available to readers. Please disclose at the submission stage any restrictions on the availability of materials or information. New methods and protocols should be described in detail while well-established methods can be briefly described and appropriately cited.

Research manuscripts reporting large datasets that are deposited in a publicly available database should specify where the data have been deposited and provide the relevant accession numbers. If the accession numbers have not yet been obtained at the time of submission, please state that they will be provided during review. They must be provided prior to publication.

Interventionary studies involving animals or humans, and other studies require ethical approval must list the authority that provided approval and the corresponding ethical approval code.

4.1. Luego vamos cambiando el texto, voy a ir cargando los datos

Descargamos y leemos los datos

```
download.file(url = 'https://geoportal.valencia.es/server/rest/services/OPENDATA/T

aparc_bic <- read_sf("data/aparcaments_bic.geojson")
cat('Nuestros datos tienen', nrow(aparc_bic), ' estancias y ', ncol(aparc_bic), 'v

## Nuestros datos tienen 2000  estancias y  4 variables.
```

Es extraño, que el Ajuntament haya decidido poner *exactamente* 2000 aparcamientos de bicicletas. Vamos a comprobar otra forma de leer los datos (me la encontré por los interneses)

```
aparc_bic <- st_read(dsn = 'https://geoportal.valencia.es/server/rest/services/OPE

cat('Nuestros datos tienen', nrow(aparc_bic), ' estancias y ', ncol(aparc_bic), 'v

## Nuestros datos tienen 4344  estancias y  4 variables.
```

ya decía yo…
Ahora vamos a descargar los ValenBisi, EMT, FGV y población por barrios.

```
valenbisi <- st_read(dsn = 'https://geoportal.valencia.es/server/rest/services/OPE

cat('Los datos de ValenBisi tienen', nrow(valenbisi), ' estancias y ', ncol(valenb

## Los datos de ValenBisi tienen 273  estancias y  12 variables.
```

```
emt <- st_read(dsn = 'https://geoportal.valencia.es/server/rest/services/OPENDATA/

cat('Los datos de EMT tienen', nrow(emt), ' estancias y ', ncol(emt), 'variables.\

## Los datos de EMT tienen 1155  estancias y  9 variables.

fgv <- st_read(dsn = 'https://geoportal.valencia.es/server/rest/services/OPENDATA/

cat('Los datos de FGV tienen', nrow(fgv), ' estancias y ', ncol(fgv), 'variables.\

## Los datos de FGV tienen 144  estancias y  8 variables.
```

Como queremos ver si existe alguna relación entre densidad demográfica y aparcamientos de bicicletas y otros medios de transporte, cargamos tambien datos poblacionales.

Notamos que en la sección de *Datos Demográficos* del portal de datos abiertos https://opendata.vlci.valencia.es/ está vacío. Vemos que existe un conjuntos de datos llamado *Manzanas catastrales con datos de población*, así que los cargamos__

```
pob_mal <- st_read(dsn = 'https://geoportal.valencia.es/server/rest/services/OPEND

cat('Los datos de población tienen', nrow(pob_mal), ' estancias y ', ncol(pob_mal)

## Los datos de población tienen 88  estancias y  7 variables.

names(pob_mal)

## [1] "objectid"             "codbarrio"            "nombre"
## [4] "coddistbar"           "coddistrit"           "gis.gis.BARRIOS.area"
## [7] "geometry"
```

Pero vemos que no hay datos de población. Por tanto tendremos que buscar otras fuentes.

```
url_descarga <- "https://www.valencia.es/estadistica/inf_dtba/2026/Barrios2026.zip
carpeta_datos <- "data"
entrada <- file.path(carpeta_datos, "Barrios2026.zip")
salida  <- "densidad_barrios_2025.csv"
```

Mediante inspección, vemos que el zip contiene la información de los 88 barrios de Valencia. Como es muy complicado automatizar esta recolección de datos de cada uno de los xlsx, utilizamos ayuda de *Claude* para extraer esa información.

```
## Usando el zip ya descargado: data/Barrios2026.zip

##   distrito barrio nombre_barrio personas superficie_ha densidad_hab_ha
## 1        1      1        la Seu     3275        22.107        148.1431
## 2        1      2      la Xerea     4277        31.165        137.2373
## 3        1      3      el Carme     7252        38.407        188.8197
## 4        1      4      el Pilar     5472        16.161        338.5929
## 5        1      5     el Mercat     4101        17.279        237.3401
## 6        1      6 Sant Francesc     6133        43.866        139.8122

barrios <- st_read(dsn = 'https://geoportal.valencia.es/server/rest/services/OPEND

cat('Los datos de barrios tienen', nrow(barrios), ' estancias y ', ncol(barrios),
```

```
## Los datos de barrios tienen 88  estancias y  7 variables.
```

```
head(barrios)
```

```
## Simple feature collection with 6 features and 6 fields
## Geometry type: POLYGON
## Dimension:     XY
## Bounding box:  xmin: -0.4025012 ymin: 39.47809 xmax: -0.3515598 ymax: 39.53302
## Geodetic CRS:  WGS 84
##   objectid codbarrio      nombre coddistbar coddistrit gis.gis.BARRIOS.area
## 1        5         1  BENIFARAIG        171         17              14228152
## 2        9         1   BENICALAP        161         16              17189645
## 3       10         2   TORREFIEL        152         15               7132926
## 4       11         4      TORMOS        054          5                279952
## 5       12         5 SANT ANTONI        055          5               2497341
## 6       13         1  BENIMACLET        141         14               7431832
##                           geometry
## 1 POLYGON ((-0.3763896 39.528...
## 2 POLYGON ((-0.3813917 39.498...
## 3 POLYGON ((-0.3712209 39.498...
## 4 POLYGON ((-0.3821647 39.490...
## 5 POLYGON ((-0.3738837 39.492...
## 6 POLYGON ((-0.3590119 39.488...
```

```
distritos <- st_read(dsn = 'https://geoportal.valencia.es/server/rest/services/OPE

cat('Los datos de distritos tienen', nrow(distritos), ' estancias y ', ncol(distri
```

```
## Los datos de distritos tienen 22  estancias y  5 variables.
```

```
head(distritos)
```

```
## Simple feature collection with 6 features and 4 fields
## Geometry type: POLYGON
## Dimension:     XY
## Bounding box:  xmin: -0.3962658 ymin: 39.42634 xmax: -0.3345913 ymax: 39.49206
## Geodetic CRS:  WGS 84
##   objectid          nombre coddistrit gis.gis.DISTRITOS.area
## 1        1 QUATRE CARRERES         10                     NA
## 2        6       LA SAIDIA          5               19439315
## 3        8 EL PLA DEL REAL          6               16927129
## 4        9    CIUTAT VELLA          1               16898516
## 5       10       EXTRAMURS          3               19716162
## 6       12      L'EIXAMPLE          2               17331405
##                           geometry
## 1 POLYGON ((-0.3359056 39.442...
## 2 POLYGON ((-0.3894709 39.488...
## 3 POLYGON ((-0.362286 39.4848...
## 4 POLYGON ((-0.3810546 39.482...
## 5 POLYGON ((-0.3833952 39.481...
## 6 POLYGON ((-0.3645362 39.471...
```

Ahora por último, lo juntamos todo en un mismo DataFrame (polígonos de distritos, barrios y densidad de población)

```r
# --- 1. Polígonos de barrio ------------------------------------------------
# Cruce por código de distrito + código de barrio (codbarrio se repite entre
# distritos, así que hacen falta los dos).
poligonos_barrios <- barrios |>
  mutate(coddistrit = as.integer(coddistrit),
         codbarrio  = as.integer(codbarrio)) |>
  select(coddistrit, codbarrio, nombre_geo = nombre)

resultado <- resultado |>
  left_join(poligonos_barrios,
            by = c("distrito" = "coddistrit", "barrio" = "codbarrio")) |>
  st_as_sf()

st_geometry(resultado) <- "geom_barrio"   # nombre claro para la geometría activa


distritos_unidos <- distritos |>
  st_make_valid() |>
  mutate(coddistrit = as.integer(coddistrit)) |>
  group_by(coddistrit) |>
  summarise(nombre_distrito = first(nombre), .groups = "drop")  # une las piezas

fila <- match(resultado$distrito, distritos_unidos$coddistrit)
resultado$nombre_distrito <- distritos_unidos$nombre_distrito[fila]
resultado$geom_distrito   <- st_geometry(distritos_unidos)[fila]   # 2ª columna de

# --- 3. Densidad del distrito (población total / superficie total) ---------
# El Port no tiene población pero su superficie cuenta en el distrito 11.
resultado <- resultado |>
  group_by(distrito) |>
  mutate(densidad_distrito_hab_ha = sum(personas, na.rm = TRUE) /
                                    sum(superficie_ha, na.rm = TRUE)) |>
  ungroup() |>
  relocate(nombre_distrito, .after = barrio)

# --- Comprobaciones --------------------------------------------------------
cat("Barrios sin polígono de barrio:  ", sum(st_is_empty(resultado$geom_barrio)),
```

```
## Barrios sin polígono de barrio:   1
```

```r
cat("Barrios sin polígono de distrito:", sum(is.na(fila)), "\n")
```

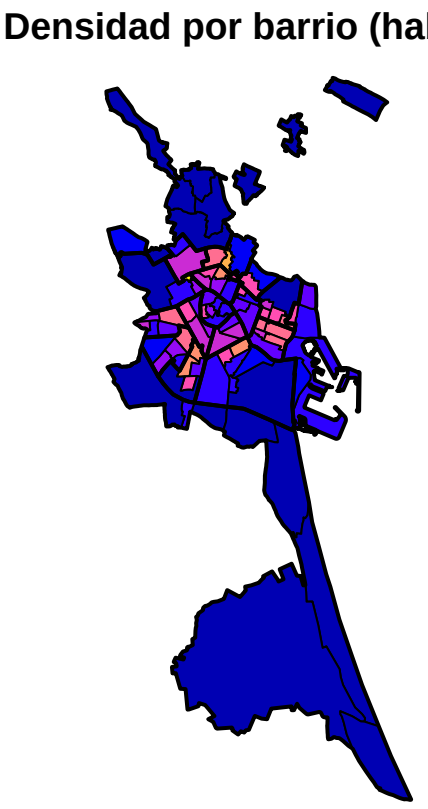
```
## Barrios sin polígono de distrito: 0
```

```r
print(resultado)
```

```
## Simple feature collection with 89 features and 9 fields
## Active geometry column: geom_barrio (with 1 geometry empty)
## Geometry type: POLYGON
## Dimension:     XY
## Bounding box:  xmin: -0.432535 ymin: 39.27893 xmax: -0.2753685 ymax: 39.56659
## Geodetic CRS:  WGS 84
## # A tibble: 89 x 11
##    distrito barrio nombre_distrito nombre_barrio    personas superficie_ha
```

```
##       <int>  <int> <chr>           <chr>               <dbl>          <dbl>
##  1        1      1 CIUTAT VELLA    la Seu               3275           22.1
##  2        1      2 CIUTAT VELLA    la Xerea             4277           31.2
##  3        1      3 CIUTAT VELLA    el Carme             7252           38.4
##  4        1      4 CIUTAT VELLA    el Pilar             5472           16.2
##  5        1      5 CIUTAT VELLA    el Mercat            4101           17.3
##  6        1      6 CIUTAT VELLA    Sant Francesc        6133           43.9
##  7        2      1 L'EIXAMPLE      Russafa             25514           87.8
##  8        2      2 L'EIXAMPLE      el Pla del Remei     7344           38.7
##  9        2      3 L'EIXAMPLE      la Gran Via         12378           46.9
## 10        3      1 EXTRAMURS       el Botànic           7033           37.0
## # i 79 more rows
## # i 5 more variables: densidad_hab_ha <dbl>, nombre_geo <chr>,
## #   geom_barrio <POLYGON [°]>, geom_distrito <POLYGON [°]>,
## #   densidad_distrito_hab_ha <dbl>
```

```
# --- Mapa de comprobación: barrios coloreados, límites de distrito encima ---
plot(resultado["densidad_hab_ha"], main = "Densidad por barrio (hab/ha)",
     reset = FALSE)
plot(st_geometry(distritos_unidos), add = TRUE, border = "black", lwd = 2)
```



5. Results

This section may be divided by subheadings. It should provide a concise and precise description of the experimental results, their interpretation as well as the experimental conclusions that can be drawn.

5.1. Subsection Heading Here

Subsection text here.

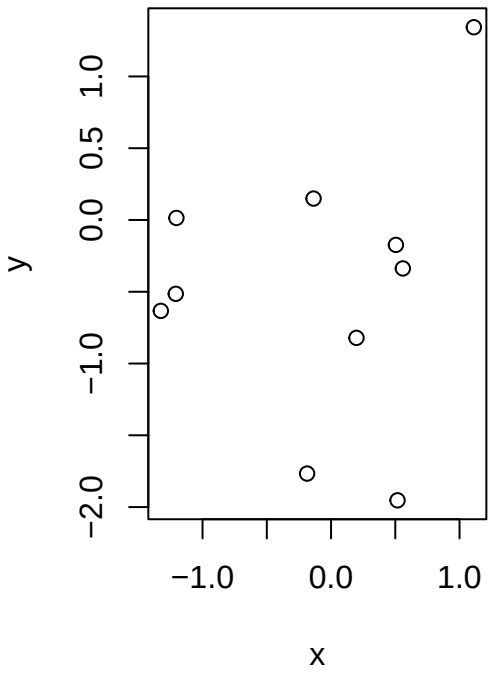


Figure 1. A figure added with a code chunk.

5.1.1. Subsubsection Heading Here

Bulleted lists look like this:

- First bullet
- Second bullet
- Third bullet

Numbered lists can be added as follows:

1. First item
2. Second item
3. Third item

The text continues here.

5.2. Figures, Tables and Schemes

All figures and tables should be cited in the main text as Figure 1, 3, etc. To get cross-reference to figure generated by R chunks include the `\\label{}` tag in the `fig.cap` attribute of the R chunk: `fig.cap = "Fancy Caption\\label{fig:plot}"`.

When making tables using `kable`, it is suggested to use the `format="latex"` and `tabl.envir="table"` arguments to ensure table numbering and compatibility with the mdpi document class.

Table 3. This is a table caption. Tables should be placed in the main text near to the first time they are~cited.

	mpg	cyl	disp
Mazda RX4	21.0	6	160
Mazda RX4 Wag	21.0	6	160
Datsun 710	22.8	4	108
Hornet 4 Drive	21.4	6	258
Hornet Sportabout	18.7	8	360

For a very wide table, landscape layouts are allowed.

Table 4. This is a very wide table

Title.1	Title.2	Title.3	Title.4
Entry 1	Data	Data	This cell has some longer content that runs over two lines
Entry 2	Data	Data	Data

5.3. Formatting of Mathematical Components

This is an example of an equation:

$$a = 1.$$

If you want numbered equations use Latex and wrap in the equation environment:

$$a = 1, \tag{1}$$

the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

This is the example 2 of equation:

$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z \tag{2}$$

Theorem-type environments (including propositions, lemmas, corollaries etc.) can be formatted as follows:

Example of a theorem:

Theorem 1. *Example text of a theorem*

The text continues here. Proofs must be formatted as follows:

Example of a proof:

Proof of Theorem1. Text of the proof. Note that the phrase "of Theorem 1'' is optional if it is clear which theorem is being referred to.

□

The text continues here.

6. Discussion

Authors should discuss the results and how they can be interpreted in perspective of previous studies and of the working hypotheses. The findings and their implications should be discussed in the broadest context possible. Future research directions may also be highlighted.

7. Conclusion

This section is not mandatory, but can be added to the manuscript if the discussion is unusually long or complex.

8. Patents

This section is not mandatory, but may be added if there are patents resulting from the work reported in this manuscript.

9. Anexo: Información de la sesión

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24)
## Platform: aarch64-apple-darwin23
## Running under: macOS Sonoma 14.7.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.d
## LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.d
##
```

```
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Madrid
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] readxl_1.5.0.1  knitr_1.52      lubridate_1.9.5 forcats_1.0.1
##  [5] stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2     readr_2.2.0
##  [9] tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0
## [13] sf_1.1-3
##
## loaded via a namespace (and not attached):
##  [1] s2_1.1.13          utf8_1.2.6         generics_0.1.4     class_7.3-23
##  [5] KernSmooth_2.23-26 stringi_1.8.9      hms_1.1.4          digest_0.6.39
##  [9] magrittr_2.0.5     evaluate_1.0.5     grid_4.6.1         timechange_0.4.0
## [13] RColorBrewer_1.1-3 fastmap_1.2.0      cellranger_1.1.0   e1071_1.7-17
## [17] DBI_1.3.0          scales_1.4.0       cli_3.6.6          rlang_1.3.0
## [21] units_1.0-1        withr_3.0.3        yaml_2.3.12        otel_0.2.0
## [25] tools_4.6.1        tzdb_0.5.0         vctrs_0.7.3        R6_2.6.1
## [29] proxy_0.4-29       lifecycle_1.0.5    classInt_0.4-11    rticles_0.27
## [33] pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6       glue_1.8.1
## [37] Rcpp_1.1.2         xfun_0.61          tidyselect_1.2.1   rstudioapi_0.19.0
## [41] farver_2.1.2       htmltools_0.5.9    rmarkdown_2.32     wk_0.9.5
## [45] compiler_4.6.1     S7_0.2.2
```

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/1010000/s1, Figure S1: title; Table S1: title; Video S1: title.

Author Contributions: For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used "X.X. and Y.Y. conceive and designed the experiments; X.X. performed the experiments; X.X. and Y.Y. analyzed the data; W.W. contributed reagents/materials/analysis tools; Y.Y. wrote the paper.'' Authorship must be limited to those who have contributed substantially to the work reported.

Funding: was funded by NAME OF FUNDER grant number XXX.'' and and `The APC was funded by XXX''. Check carefully that the details given are accurate and use the standard spellin` `of funding agency names at funding. Please add:`This research received no external funding'' or "This research https://search.crossref.org/funding, any errors may affect your future

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Sample Availability: Samples of the compounds ..... are available from the authors.

Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronym
LD	linear dichroism

Appendix J

Appendix J.1

The appendix is an optional section that can contain details and data supplemental to the main text. For example, explanations of experimental details that would disrupt the flow of the main text, but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data is shown in the main text can be added here if brief, or as Supplementary data. Mathematical proofs of results not central to the paper can be added as an appendix.

Appendix K

All appendix sections must be cited in the main text. In the appendixes, Figures, Tables, etc. should be labeled starting with 'A', e.g., Figure A1, Figure A2, etc.

References

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